

Univessel SU 2 L - 0.6-2 L working volume

sartorius-univessel-su-2

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	600 mL	Source-supported
Maximum working volume	2 L	Source-supported
Nominal vessel / bag size	2 L	Source-supported
Total vessel volume loaded into the form	2.6 L	Source-supported
Initial working volume loaded into the form	600 mL	Derived / estimated
Product-contact material	Polycarbonate vessel; silicone / C-Flex tubing; EPDM seal	Source-supported
Height : diameter	1.16	Derived / estimated
Internal / chamber diameter	0.13 m	Source-supported
Liquid height at maximum working volume	0.1507 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	App assumption
Impeller count	2	Source-supported
Impeller diameter	0.054 m	Source-supported
Power number	0.4	App assumption
Initial agitation	100 rpm	App assumption
Maximum agitation	250 rpm	App assumption
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	No	App assumption
Maximum pressure	0.5 bar(g)	Source-supported
Default operating pressure	0.05 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 600 mL

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 600 mL.

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

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[Manufacturer publication](#)

Maximum working volume - 2 L

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2 L.

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

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Nominal vessel / bag size - 2 L

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2 L.

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

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Total vessel volume loaded into the form - 2.6 L

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.6 L.

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

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Initial working volume loaded into the form - 600 mL

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(600 mL, 2 L) = 600 mL

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Polycarbonate vessel; silicone / C-Flex tubing; EPDM seal

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Polycarbonate vessel; silicone / C-Flex tubing; EPDM seal.

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

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Height : diameter - 1.16

Derived / estimated | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 0.002 \text{ m}^3$; $D = 0.13 \text{ m}$; $H = 0.15067923606333286 \text{ m}$; $H/D = 1.159071046641022$

Limitations: Derived liquid aspect ratio. Does not reuse an overall vessel ratio or generic profile ratio as if it described the liquid.

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Internal / chamber diameter - 0.13 m

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.13 m.

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

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Manufacturer publication

Liquid height at maximum working volume - 0.1507 m

Derived / estimated | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 0.002 \text{ m}^3$; $D = 0.13 \text{ m}$; $H = 0.15067923606333286 \text{ m}$; $H/D = 1.159071046641022$

Limitations: Ideal-cylinder estimate from published diameter and working volume. Ignores tapered walls, curved bottoms and internals; not a manufacturer liquid-height specification.

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Manufacturer publication

Impeller / mixing type - Pitched-blade turbine

App assumption | Sartorius Univessel SU 2 L brochure - PDF page 4; Agitation

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: Actual hardware: two 3-blade segment impellers, 30 degrees, downflow. Pitched-blade is a model approximation.

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Manufacturer publication

Impeller count - 2

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

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Manufacturer publication

Impeller diameter - 0.054 m

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.054 m.

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

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Manufacturer publication

Power number - 0.4

App assumption | Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014 - PDF page 5; Process engineering characterization, printed p. 18

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0.4.

Limitations: De Wilde 2014 PDF p. 5 reports torque-based Ne about 1.3 for the TWO three-blade segment impeller configuration in turbulent flow ($Re > 10000$), not an unambiguous per-impeller coefficient. Retained app coefficient is unverified; do not silently apply configuration-level Ne to each impeller or transfer it to mixed impellers.

Local original: Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014 - PDF page 5

Initial agitation - 100 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseRpm = 100`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 250 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxRpm = 250`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxVvm = 0.3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - No

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].baffled = false

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

Source-supported | Sartorius Univessel SU 2 L brochure - PDF page 4; Technical Data

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.5 bar(g).

Limitations: Diameter is the inner top diameter; wall has 1.5 degree slope. Maximum operating pressure is 0.5 bar(g). Do not substitute the heating-jacket rating.

Local original: [Sartorius Univessel SU 2 L brochure - PDF page 4](#)

Manufacturer publication

Default operating pressure - 0.05 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].operatingPressure = 0.05

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

Reviewed De Wilde et al., Superior Scalability of Single-Use Bioreactors, BPI September 2014, PDF pp. 1-6 (printed 14-19).

Manufacturer-authored characterization article, not a current hardware datasheet. Archived source:

[sartorius-dewilde-scalability-2014-27b44451dc52.pdf](#)

2014 Table 1 corroborates 130 mm vessel and 54 mm impeller diameters, but states 1 L minimum and 3 L total versus selected datasheet 0.6 L and 2.6 L. Keep the selected datasheet values. Historical liquid height 177 mm, vessel height 240 mm and spacing 70 mm remain unapplied pending configuration equivalence; current liquid height remains a cylinder estimate.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Supplementary archived source: [Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014 - PDF page 2](#)

Historical comparison only; not blanket validation of current model

Local original document identities

Sartorius Univessel SU 2 L brochure

[sartorius-univessel-su-2-l-brochure-ea5699ea1bd5.pdf](#)

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Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014

[sartorius-dewilde-scalability-2014-27b44451dc52.pdf](#)

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